

Weekly Internship Report (23/5/26 - 29/5/26)

WEEK 2

Intern's Name: Atharv Gupta

Internship Week: Week 2 (23/5/26 - 29/5/26)

Company/Organization Name: Red Kross Research Foundation

Mentor's Name: Shivam Mittal

Week Overview:

- Learned MD5 hash code information extraction.
- Studied Threat Intelligence concepts such as IR, Access Control, Pyramid of Pain, CVE, TTPs, Threat Actors, and Cybercrime categories.
- Explored Docker, Docker Networking, and Kubernetes basics.
- Learned the ransomware attack process.
- Received an introduction to OSINT and CSINT.
- Study the Threat Intelligence Framework.
- Conducted research on OSINT, CSINT, and Google Hacking Database (GHDB).
- Learn the working and usage of Burp Suite and Foxy Proxy.
- Performed Web Vulnerability Analysis on XXE Injection using Burp Suite.

Learning Outcomes:

- Gained knowledge of Cyber Threat Intelligence concepts and frameworks.
- Understood the basics of Docker and Kubernetes.
- Learned ransomware attack methodologies.
- Improved understanding of OSINT and CSINT techniques.
- Acquired practical exposure to Burp Suite and web security testing.
- Learned how XXE vulnerabilities can be identified and analyzed.

Challenges Faced:

- Understanding Threat Intelligence terminology and frameworks.
- Learning containerization concepts such as Docker and Kubernetes.
- Analyzing web vulnerabilities using security testing tools.

Goals for Next Week:

- Learn OSINT techniques and intelligence gathering methodologies.
- Explore practical applications for OSINT in cybersecurity investigations.
- Study the role of Artificial Intelligence in Penetration Testing.
- Understand how AI can assist in vulnerability assessment and security testing.
- Continue improving cybersecurity research and documentation skills.

Mentor Feedback and Suggestions:

- Good progress in cybersecurity research and learning activities.
- Continue improving practical skills and documentation practices.
- Maintain regular participation and consistency in assigned tasks.

Additional Notes:

- This week enhanced my understanding of Threat Intelligence, OSINT, web security testing, and container technologies.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies
2	Threat Intelligence, Container Security and Web Security Research	Studied MD5 hash extraction, Threat Intelligence concepts, Docker and Docker Networking, Kubernetes basics, Ransomware workflow, OSINT and CSINT, Threat Intelligence Framework, GHDB research, Burp Suite and FoxyProxy, and performed XXE Injection	Learned threat intelligence concepts including CVE, TTPs, Pyramid of Pain, threat actors, OSINT/CSINT methodologies, container technologies, web vulnerability assessment, and XXE exploitation techniques	Enhanced practical cybersecurity research, threat analysis, reconnaissance, and web application security skills

		vulnerability analysis using Burp Suite		
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